

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 4: Convolution, System Properties, and Difference Equations

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1 Impulse Response and the Convolution Sum

$$h[n] = H\{\delta[n]\}. \quad (\text{Mitra } \S 4.3)$$

Fact.

LTI systems with zero initial conditions (ICs) are **completely characterized** by $h[n]$.

Proof of fact. Write

$$x[n] = \sum_k x[k] \delta[n - k]^1 \quad (\text{Mitra } 4.15)$$

$y[n] = H\{x[n]\}$. If $H\{\cdot\}$ is LTI, then

$$\begin{aligned} y[n] &= H\left\{\sum_k x[k] \delta[n - k]\right\} \\ &= \sum_k x[k] H\{\delta[n - k]\} && \text{(linearity)} \\ &= \sum_k x[k] h[n - k]. && (\text{Mitra } 2.20a, 4.16a) \quad \blacksquare \end{aligned}$$

The summation

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] \quad (\text{Mitra } 2.20a, 4.16a)$$

is called the **convolution sum** of the sequences $x[n]$ and $h[n]$:

$$y[n] = x[n] \circledast h[n]. \quad (\text{Mitra 2.21})$$

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Example (Computing the Convolution Sum).

We compute $y[n] = x[n] \circledast h[n]$ numerically for a short input and a short impulse response. Here $x[n]$ is the input and $h[n] = H\{\delta[n]\}$ is the impulse response of an LTI system with zero ICs. By definition,

$$y[n] = \sum_{m=-\infty}^{\infty} x[m] h[n-m]. \quad (\text{Mitra 2.20a, 4.16a})$$

Equivalently, flip $h[n]$ to form $g[m] \triangleq h[-m]$, then slide $g[m]$ relative to $x[m]$: at each n , multiply the overlapping samples and add.

Let $h[n] = \{1, 1, 1\}$ and $x[n] = \{1, 2, 3\}$, with $n = 0$ at the first listed sample of each sequence.

Flip $h[n]$ to $g[m] = h[-m]$; align $x[m]$ and $g[m]$. At $n = 0$:

m	-1	0	1	
$x[m]$	1	2	3	
$g[m]$	1	1	0	
$x[m]g[m]$	1	2	0	$\Rightarrow y[0] = 3$

Shift $g[m]$ one sample to the right to compute $y[1] = 6$, and continue for each n . Then

$$y[0] = \sum_{m=-\infty}^{\infty} x[m] h[0-m] = \sum_{m=-\infty}^{\infty} x[m] g[m], \quad g[m] \triangleq h[-m].$$

¹Following Mitra (§4.3): in $\sum_k x[k] \delta[n-k]$, the index k labels the k th sample $x[k]$ of $\{x[n]\}$. In Eqs. (4.16a)–(4.16b), k is the convolution summation index; Eq. (4.16b) forms $y[n]$ as a sum of input samples $x[n-k]$ weighted by $h[k]$ (see Mitra, Fig. 4.7).

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With

$$g[m] = \{\dots, 0, 1, 1, 1, 0, 0, \dots\}, \quad x[m] = \{\dots, 0, 1, 2, 3, 0, \dots\},$$

with $m = 0$ at the samples 1 and 2, respectively.

$$y[0] = 0 + 1 + 2 + 0 + 0 = 3, \quad y[1] = \sum_{m=-\infty}^{\infty} x[m] g[m-1] = 6.$$

Continuing in this way,

$$y[n] = \{\dots, 0, 1, 3, 6, 5, 3, 0, \dots\}, \quad n = 0 \text{ at the sample 3.}$$

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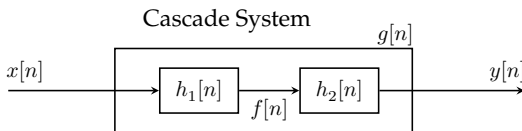
1.1 Convolution Properties

Commutative: $x[n] \otimes h[n] = h[n] \otimes x[n].$ (Mittra 4.17)

Associative: $(x[n] \otimes h[n]) \otimes y[n] = x[n] \otimes (h[n] \otimes y[n]).$ (Mittra 4.18)

Distributive: $x[n] \otimes (h[n] + y[n]) = x[n] \otimes h[n] + x[n] \otimes y[n].$ (Mittra 4.19)

1.2 Cascade System

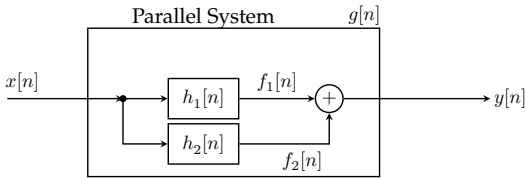


$$\begin{aligned} f[n] &= x[n] \otimes h_1[n], \\ y[n] &= f[n] \otimes h_2[n] \\ &= (x[n] \otimes h_1[n]) \otimes h_2[n] \end{aligned}$$

$$y[n] = x[n] \otimes g[n], \quad g[n] = h_1[n] \otimes h_2[n].$$

(Mittra 4.28)

4 1.3 Parallel System



$$y[n] = f_1[n] + f_2[n],$$

$$f_1[n] = x[n] \otimes h_1[n],$$

$$f_2[n] = x[n] \otimes h_2[n],$$

$$y[n] = x[n] \otimes h_1[n] + x[n] \otimes h_2[n]$$

$$y[n] = x[n] \otimes g[n], \quad g[n] = h_1[n] + h_2[n].$$

(Mitra 4.31)

5 1.4 $h[n]$ and Causality

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] \quad (\text{Mitra 2.20a, 4.16a})$$

$$= \sum_{k=-\infty}^{\infty} x[n-k] h[k]. \quad (\text{Mitra 2.20b, 4.16b})$$

Causality.

$y[n]$ is independent of $x[m]$ for all $m > n \iff h[n] = 0$ for all $n < 0$.² An LTI discrete-time system is **causal** if and only if its impulse response $h[n]$ is a “causal” sequence, i.e., $h[n] = 0$ for all $n < 0$ (Mitra 4.27).

²The handout uses $\Rightarrow$; here we use $\Leftrightarrow$ since both directions hold for LTI systems.

Example.

The FIR system below uses only present and past inputs ($k \geq 0$):

$$\begin{aligned}
 y[n] &= \sum_{k=0}^M \alpha_k x[n-k] \\
 \Rightarrow h[n] &= \sum_{k=0}^M \alpha_k \delta[n-k] \\
 &= \begin{cases} \alpha_n, & 0 \leq n \leq M \\ 0, & \text{else} \end{cases}
 \end{aligned}$$

Since $h[n] = 0$ for all $n < 0$, the system is causal: $y[n]$ depends only on $x[m]$ with $m \leq n$. (Mittra 4.14)

6 1.5 $h[n]$ and Stability

BIBO Stability³.

An LTI discrete-time system is **BIBO stable** if and only if its impulse response $\{h[n]\}$ is **absolutely summable**, i.e.,

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty. \quad (\text{Mittra 4.20, 2.42})$$

Proof (sufficiency). Assume $\{x[n]\}$ is bounded, $|x[n]| \leq B_x < \infty$ for all n , where B_x is a finite bound on the input magnitude; then the output amplitude at

³cf. Mittra Sec. 4.4.3.

time instant n , from Eq. (4.16b), is

$$\begin{aligned}
 |y[n]| &= \left| \sum_{k=-\infty}^{\infty} h[k] x[n-k] \right| \\
 &\leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]|^4 \\
 &\leq B_x \sum_{k=-\infty}^{\infty} |h[k]| \\
 &= B_x S < \infty.
 \end{aligned}
 \tag{Mitra 2.20b, 4.16b, 4.21}$$

So if

$$S = \sum_{k=-\infty}^{\infty} |h[k]| < \infty, \tag{Mitra 4.20, 2.42}$$

then $|y[n]| \leq B_y = B_x S < \infty$ for all n ; hence absolute summability is a **sufficient** condition for BIBO stability. ■

7 Necessity. Let $x[n] = \text{sgn}(h[-n])$, (Mitra 4.22) where for real-valued α ,

$$\text{sgn}(\alpha) = \begin{cases} +1, & \alpha > 0 \\ -1, & \alpha < 0 \\ 0, & \alpha = 0. \end{cases}$$

Then

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k] = \sum_{k=-\infty}^{\infty} h[k] \text{sgn}(h[k-n]).$$

At $n = 0$,

$$y[0] = \sum_{k=-\infty}^{\infty} h[k] \text{sgn}(h[k]) = \sum_{k=-\infty}^{\infty} |h[k]| = S. \tag{Mitra 4.23}$$

⁴The summation $\sum_k$ adds all terms $h[k]x[n-k]$. By the triangle inequality, the absolute value of a sum is at most the sum of absolute values, so $|\sum_k h[k]x[n-k]| \leq \sum_k |h[k]x[n-k]|$.

So for $y[0]$ to be finite (which is necessary for $y[n]$ to be bounded), it is **necessary** that $\sum_{k=-\infty}^{\infty} |h[k]| < \infty$. ■

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Example (Mitra Ex. 4.16).

Consider a causal LTI discrete-time system with an impulse response given by $h[n] = \alpha^n u[n]$. For this system,⁴

$$\begin{aligned}
 S &= \sum_{n=-\infty}^{\infty} |h[n]| \\
 &= \sum_{n=-\infty}^{\infty} |\alpha^n u[n]| \\
 &= \sum_{n=-\infty}^{\infty} |\alpha^n| u[n] \\
 &= \sum_{n=0}^{\infty} |\alpha|^n \\
 &= \frac{1}{1 - |\alpha|} \quad \text{for } |\alpha| < 1. \quad ^5
 \end{aligned}$$

Therefore, $S < \infty$ if $|\alpha| < 1$, for which the above system is BIBO stable. On the other hand, if $|\alpha| \geq 1$, the infinite series $\sum_{n=0}^{\infty} |\alpha|^n$ does not converge, and the above causal system is not BIBO stable.

⁴Since $|ab| = |a||b|$ and $u[n] \in \{0, 1\}$, we have $|u[n]| = u[n]$ and thus $|\alpha^n u[n]| = |\alpha^n| u[n]$.

⁵The last step applies the geometric series formula, which gives a finite sum when the common ratio has magnitude less than one.

Important Facts.

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad |a| < 1 \quad (\text{Mitra Ex. 4.6})$$

$$\sum_{n=0}^{N-1} a^n = \begin{cases} \frac{1-a^N}{1-a}, & a \neq 1 \\ N, & a = 1. \end{cases}$$

9 2 Linear Difference Equations

$$\sum_{k=0}^N d_k y[n-k] = \sum_{q=0}^M p_q x[n-q], \quad (\text{Mitra 4.32})$$

finite order $\triangleq \max(N, M)$.

If we assume causality, then $y[n]$ can be computed for all $n \geq n_0$ from $x[n]$ for $n \geq n_0$ and the ICs

$$y[n_0-1], y[n_0-2], \dots, y[n_0-N], \quad x[n_0-1], x[n_0-2], \dots, x[n_0-M]. \quad (\text{Mitra 4.33})$$

$$y[n] = -\sum_{k=1}^N \frac{d_k}{d_0} y[n-k] + \sum_{q=0}^M \frac{p_q}{d_0} x[n-q]. \quad (\text{Mitra 4.33})$$

- **FIR** (finite impulse response): $N = 0, M \geq 0$.
- **IIR** (infinite impulse response): $N > 0, M \geq 0$.

10 2.1 FIR Systems

$$y[n] = \sum_{q=0}^M \tilde{p}_q x[n-q] \implies h[n] = \begin{cases} \tilde{p}_n, & n = 0, \dots, M \\ 0, & \text{else.} \end{cases} \quad (\text{Mitra 4.14})$$

Generically called “moving average” filters:

- Moving average: $y[n] = \frac{x[n] + x[n-1]}{2}$. (Mitra 4.4)
- Finite difference: $y[n] = x[n] - x[n-1]$.

- Simple echo: $y[n] = x[n] + \alpha x[n - M]$.

2.2 IIR Systems

$$y[n] = \sum_{k=1}^N \tilde{d}_k y[n - k] + \sum_{q=0}^M \tilde{p}_q x[n - q]. \quad (\text{Mitra 4.33})?$$

- $M = 0$: **autoregressive (AR)**.
- $M > 0$: **autoregressive moving average (ARMA)**.
- Recursive (feedback loop); harder to characterize:
 - Requires an IC (infinitely many solutions for $y[n]$ given only $x[n]$ with unspecified ICs).
 - Harder in general to compute the impulse response.

11 2.3 Example: $y[n] = a y[n - 1] + x[n]$

What is the impulse response?

1. Zero ICs ($y[n] = 0$ for all $n < 0$).

$$\begin{aligned} y[0] &= a y[-1] + \delta[0] = 1, \\ y[1] &= a y[0] + \delta[1] = a, \\ y[2] &= a y[1] + \delta[2] = a^2, \\ &\vdots \\ y[n] &= a^n, \quad n \geq 0 \implies h[n] = a^n u[n]. \end{aligned} \quad (\text{Mitra Ex. 4.16})$$

2. $y[n] = 0$ for all $n \geq 0$. Note $y[n] = a y[n-1] + x[n] \Rightarrow y[n-1] = a^{-1}(y[n] - x[n])$, so

$$\begin{aligned} y[-1] &= a^{-1}(y[0] - \delta[0]) = -a^{-1}, \\ y[-2] &= a^{-1}(y[-1] - \delta[-1]) = -a^{-2}, \\ &\vdots \\ y[n] &= -a^n, \quad n \leq -1 \implies h[n] = -a^n u[-1 - n]. \end{aligned}$$

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2.4 Why?

If we have a particular solution $y_p[n]$ that satisfies a given LDE, then any solution $y_h[n]$ to the “homogeneous LDE” ($x[n] = 0$) gives another valid solution of the form

$$y[n] = y_p[n] + y_h[n], \quad (\text{Mitra 4.39})$$

$$\text{Homogeneous LDE: } \sum_{k=0}^N d_k y[n-k] = 0. \quad (\text{Mitra 4.40})$$

Homogeneous LDE.

$y[n] = a y[n-1] \Rightarrow$ no unique solution! $y[n] = A a^n$ is a solution for any A .
Particular solutions:

$$(1) \quad h[n] = a^n u[n],$$

$$(2) \quad h[n] = -a^n u[-1-n].$$

General solution: $h[n] = a^n u[n] + A a^n$.

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Using $u[-1-n] = 1 - u[n]$,

$$-a^n(1 - u[n]) = a^n u[n] - a^n.$$

Consider candidate solutions to a generic homogeneous equation $\sum_{k=0}^N a_k y[n-k] = 0$ of the form $y[n] = A_\ell \lambda_\ell^n$. For what choices of A_ℓ and λ_ℓ is this a valid solution?

$$\sum_{k=0}^N a_k A_\ell \lambda_\ell^{(n-k)} = 0$$

$$A_\ell \lambda_\ell^n \sum_{k=0}^N a_k \lambda_\ell^{-k} = 0$$

Note: $\sum_{k=0}^N a_k z^{-k}$ is a polynomial in z^{-1} . In the case of simple roots, the homo-

geneous equation has solutions

$$y[n] = \sum_{\ell=1}^N A_{\ell} \lambda_{\ell}^n, \quad (\text{Mitra 4.43})$$

where λ_{ℓ} are the roots of the characteristic polynomial (Mitra §4.6.1) and A_{ℓ} are arbitrary for $\ell = 1, \dots, N \Rightarrow$ infinitely many solutions.

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2.5 Inverse Filter

$$x[n] \rightarrow \boxed{h_1[n]} \rightarrow y[n] \rightarrow \boxed{h_2[n]} \rightarrow x[n]$$

Examples: $h_1[n]$ channel, $h_2[n]$ equalizer; $h_1[n]$ is blur, $h_2[n]$ is deblurring. In either case,

$$h_1[n] \otimes h_2[n] = \delta[n]. \quad (\text{Mitra 4.29})$$

Example.

$y[n] = a y[n-1] + x[n]$ (IIR), with $h_1[n] = a^n u[n]$. Then

$$x[n] = y[n] - a y[n-1] \Rightarrow \text{FIR}, \quad h_2[n] = \delta[n] - a \delta[n-1] \quad (\text{inverse filter}).$$

Caution: ICs should be treated carefully — the inverse filter may not be causal or stable.

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 3.2, 3.3
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 2.5, 2.6